

PE6201

Emerging AI Technologies

Class 1 — The Landscape

Ajay Vikram Singh · Group C Mon 3 Aug | Group B Tue 4 Aug · 1830–2145

MSc in Enterprise Artificial Intelligence





Ajay Vikram Singh

Course author and lead instructor. 20+ years building products — 10+ of them in AI.

Meta

Led the AI recommendation systems behind Facebook, Instagram and Reels — billions of recommendations a day, to over two billion people.

Nokia

Built the AI-as-a-Service business from zero to scale, with global enterprise customers. E.g. We built systems that saved 100s of millions of KWHs of energy yearly (like driving an EV 50k times around the earth or charging 10s of billions of iPhones 0% to 100%)

IMDA Singapore

Helped run the national AI plan for small and medium enterprises.

Praana Health

Chief Product & Technology Officer — clinical-grade healthcare at scale. Using AI to help people live healthy, longer.

M.Tech AI, NUS · Stanford AI Professional Program · Masters Public Administration, NUS ·
MBA, ISB/India · B.Tech- Electrical Engineering IIT/India



I have built products for twenty years — ten of them in AI.
This course teaches it the way it is built in real life.

By the end of six weeks you will have built with AI — without code, and with code. Not read about it. Built with it, on a problem you chose.

Two-thirds of you have no technical background.
That is exactly who this course is designed for.

One-third of you do.
You are not here to be bored. There is depth in every session — and in week four, your team will need you.



Where this course sits

Twelve courses, 30 academic units, three trimesters. This is the first.

T1 — FOUNDATIONS 9 AU	T2 — ENTERPRISE & GOVERNANCE 10.5 AU	T3 — LEADERSHIP & CAPSTONE 10.5 AU
PE6201 Emerging AI Technologies ← you are here PE6202 AI Automation & Application in Enterprise PE6203 Generative AI and Agentic AI PE6204 AI-Driven Cybersecurity	PE6205 Enterprise AI & Analytics PE6206 Data Architecture, Management & Governance PE6207 AI Ecosystems: Platforms, Integration, Tools PE6208 Responsible AI and Governance	PE6209 Leading AI-driven Innovation & Transformation PE6210 Multimodal AI and Emerging Use Cases PE6211 Agile AI Product Development PE6212 AI Transformation Capstone

Everything else in this degree assumes what we do in these six weeks. PE6201 is also a formal prerequisite to AI Ecosystems (PE6207) and Agile AI Product Development (PE6211).



The six weeks

Each session is three hours and fifteen minutes. Today is week one.

1	The landscape	What kinds of AI exist, and how each fails
2	The modern AI stack	How any of it becomes a product
3	Foundation models and prompting	What is inside, and how to steer it
4	Agents and tool use	Systems that act, not just answer
5	Strategy and economics	Where AI creates value, and what it costs
6	Responsible AI and the frontier	How it fails, and what comes next

Every session rhythm: pre-readings before, a live build during, a one-page recap after.



Today

Five blocks. Two of them are you working, not me talking.

Opening	20 min	Where you are, how you are graded
1 · Your intuition is wrong	24 min	A poll, an argument at your table, and eight surprises
2 · Sorting vs. making	24 min	The oldest split in AI — with live code
BREAK	10 min	
3 · What scale changed	30 min	What a foundation model actually is — with live code
4 · Moderna	42 min	A real company, a real decision, and your discussion
BREAK	10 min	
5 · Choosing	30 min	The framework you leave with

Evening layout — two 10-minute breaks instead of one long one, easier after a workday.



How you are assessed

Four components. No final exam.

Foundations Build	16%	Individual	Wks 1–3 Rel W2 End Due W4 End
Applied AI System	20%	Team of 6	Wks 4–6 Rel W4 End Due W6 End
Class Participation	20%	Individual	All 6 sessions
End-of-Course Project	44%	Individual	Self-defined 1-page Problem statement Due W3 End Project Due W7 End

Nothing is due for two weeks. The first thing you submit is A1, at the end of week four.



The two assignments

One you do alone. One you do with the people at your table.

A1 · Foundations Build

16%

INDIVIDUAL

- Build a classifier and a prompt for the same task — measure both
- Build a small retrieval system, then break it on purpose
- Write a prompt, build ten test cases, improve it

Released Sun 16 Aug · Due Sun 30 Aug

A2 · Applied AI System

20%

TEAM OF 6

- Build an agent that uses at least one external tool
- Model what it costs to run at scale
- Write a responsible-deployment checklist — 10+ test cases

Released Sun 30 AUG · Due 13 Sep

Half the marks in A1 are for your reasoning, not your code. Starter notebooks are provided for every part.



Class participation is 20%.

Here is exactly how it works.

30%	Preparation and presence	You turn up, on time, having done the reading.
40%	Active engagement	You contribute.
30%	Application and synthesis	You connect ideas across weeks.

Contributing is not only speaking in front of sixty people. It counts when you answer a poll, report back for your table, build on someone else's point, or email me a question after class, start a discussion in NTULearn or participate in one

If English is not your first language, you are not at a disadvantage in this course.

Your table

You have been seated deliberately. Here is why.

Mixed on purpose

Your table mixes nationalities and backgrounds. Nobody learns much from a room of people who already agree with them.

These become your A2 team

In week four you will build something together, for 20% of your grade. It goes better if you already know each other.

You will use them today

Twice in the next hour, I will ask you to argue with the person next to you.

Right now — 60 seconds. Name, where you are from, what you studied. Everyone at the table.



Practical things

1

Everything lives on NTULearn

Slides, readings, notebooks, recaps — posted the same evening as each class.

2

Sowmiya Thirumeni is your TA

Bioengineering at NTU. She has built LLM pipelines turning research PDFs into structured data, programmed autonomous robots with computer vision, and a wearable EEG system that detects driver fatigue. She marks alongside me and is here between sessions.

3

Pre-readings before each class

Five for today. This week was the heaviest — later weeks are lighter.

4

Reach me directly

ajay.ai.education@gmail.com. If you are stuck, email early rather than late.

There is no such thing as a question too basic. If you are wondering, six other people are too.



POLL · JUST TO GET STARTED

Have you used an AI tool in the past week?

- A** Yes, most days
- B** Yes, once or twice
- C** No
- D** Not sure whether what I used counts

Wooclap results

Scan the code. This one is just to check everyone can connect.

